

Chi-Wei HU

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Aspiring software engineer currently pursuing a MS in Computer Science at USC. Proficient in C#, C++ and Python, with project experience in object-oriented programming, database management, and front-end service development. Actively seeking a software engineer position.

Education

University of Southern California(USC)

Los Angeles, U.S.

MS in Computer Science

Sept. 2024 - Present

- Coursework: Multimedia System Design (CSCI576), Foundation of Artificial Intelligence (CSCI561), Machine Learning for Data Science (CSCI552), Operating System (CSCI402), Analysis of Algorithm (CSCI570), Database Systems (CSCI585)

National Taiwan University (NTU)

Taipei, Taiwan

MS in Biochemistry

Sept. 2020 - July2022

- Thesis: Engineering of A Multi-cell Type Immunotherapy leveraging synNotch-driven Macrophages
- GPA: 3.88/4.3

National Cheng Kung University (NCKU)

Tainan, Taiwan

BA in Life Science

Sept. 2017-July 2020

Skills

- **Languages:** C#, C++, C, Python, Rust, Java, HTML, SQL
- **Databases:** MySQL, MongoDB
- **Version Control & Code Review:** Git, Mercurial, Phabricator/Diff
- **Framework:** WPF, Express.js, Flask, SwiftUI
- **General:** Object-Oriented Programming, Backend Software Engineering
- **Others:** Apache Thrift (RPC framework), Bash/ Linux OS, CI/CD, Digital Signal Processing (DCT/IDCT), Image Processing Algorithms (JPEG compression), Firmware, RS-232/RS-485 Communication protocol, SEMI Equipment Communication Standard (SECS) Communication protocol, CSS, Automation process, Embedded system, iOS development, Data Structure, Multithreaded programming, Synchronization, Kernel programming (C)

Experience

Software Engineer Intern, WhatsApp: Small Business & CoExistence

Menlo Park, California

Meta Inc.

May 2025- Aug. 2025

- Built a Rust-based CLI debugging tool on top of the Apache Thrift framework, enabling cross-language backend APIs to analyze records from ZippyDB and improving issue triaging
- Establish webhook replay functionality in message echo workflow of WhatsApp Business CloudAPI platform to ensure reliable cross-system message delivery

Software Engineer, Department of Computer-Integrated Manufacturing (CIM)

Tainan, Taiwan

Strong Electro-Mechanical Co., Ltd

Sept. 2022- May 2024

- Developed a Standalone Automation Process for etching machines in C#, utilizing SECS for device communication and improving process efficiency by 20%
- Established a C++ monitoring system for exposure machines, enabling communication with embedded systems like Arduino and DAQ devices and integrating with MySQL, resulting in a 15% reduction in downtime

Projects & Research Experience

ParkScreen — Multimodal Parkinson's Screening Tool ([Github](#)) ([Link](#))

Python, scikit-learn, Claude API, Docker

- Built with Claude: Life Sciences Hackathon, 2026 (Anthropic × Gladstone × Cerebral Valley)
- A clinical decision-support tool that screens for Parkinson's disease from three task-matched user recordings
- Fused per-channel logistic regressions on vocal and facial-AU signals via score-level late fusion (AUC 0.758, subject-level LOSO on 95 NeuroVoz subjects), then piped the fused scores into Claude Opus 4 to generate structured clinical-style reports

Multimodal Video Content Segmentation System ([Github](#))

Python, CLIP, Whisper, Ollama, TransNetV2

- Architected a multi-modal pipeline that temporally segments videos into content vs. non-content regions by fusing visual, speech, and LLM signals; scored full marks on all final test cases
- Owned the fusion engine: weighted voting across modalities, dual-path (speech vs. non-speech) classification, and context-aware confidence calibration

Operating Systems Kernel Project (Weenix) ([BitBucket](#))

C, Kernel, VFS, Concurrency

- Implemented core OS subsystems in C, including process scheduling and virtual memory management, with correct synchronization across kernel threads
- Developed system calls, pathname resolution, and VFS interfaces, while debugging concurrency issues (race conditions, deadlocks, memory corruption) in Ubuntu 16.04